Verify, Look Again, Remember: Fail-Closed Vision-Language Checks, Active Look-Around and Safety-Monotone Mission Memory for Relief-Supply Delivery Robots

Anonymous Authors

Code, data and every trial log: https://github.com/fe2131-art/rescue-mind

***Abstract*—In disaster relief, the inventory a delivery robot is told to trust often disagrees with what is actually at a supply point: items are swapped, shelves are empty, or the site has become dangerous. We present RescueMind, a mobile delivery robot that stops at the supply station and checks the item with a vision-language model (VLM) that runs on a local GPU, with no cloud service, before it carries anything to a shelter. The check is built to fail closed. The VLM never generates free text for the decision. Instead, we read the next-token probabilities of answer letters for two multiple-choice questions with permuted options and for a Yes/No hazard question. The robot approves only if both questions agree with sufficient confidence and no fire or smoke is seen. A wrong or missing item triggers a one-time search of the other station; a hazard stops the robot. Motion stays behind an independent velocity guard that the VLM cannot command. Every closed-loop trial is audited against simulator ground truth and a ROS bag. The held-out image tests and the closed-loop campaigns were pre-registered, and every amendment is published. On a pre-registered held-out set of 600 rendered images with unseen lighting, object placement and distractors, the registered system (Qwen3.5-9B) approved 5 of 408 images that should have been refused (95% CI [0.5, 2.8]%) and 187 of 191 that should have been approved. The supply check alone made 0 false approvals in 332 must-refuse images. Every false approval was a fire only partly visible behind the supply. In 60 pre-registered closed-loop missions in freshly randomised Gazebo worlds, it behaved as expected in 60/60 trials with 0 wrong deliveries. A blind-delivery baseline would have made 40. To close the remaining gap, the robot also looks at the station from 0.4 m to either side before it leaves. In a paired, pre-registered campaign on new worlds with small, partly hidden fires, this reduced deliveries despite a perceivable fire from 11 to 1 of 16 (exact McNemar $p = 0.0019$), with no false stops. A robot that resupplies the same shelter repeatedly can also remember what it saw. We let memory only reorder where the robot looks and add caution, never remove a check. We show that this bounds the per-mission probability of an unsafe delivery by the per-check false-approval rate, however stale the memory is. In a pre-registered campaign of TBD paired missions in worlds that change between missions, the registered claim (memory's speed-up without its safety cost) was TBD. With relocated supplies, the median mission time went from TBD s to TBD s. Unsafe deliveries were TBD for a naive memory that trusts what it saw and TBD for ours. Contrastive zero-shot baselines were either unsafe (SigLIP: 44/408 false approvals) or unusable (CLIP: 2/191 correct approvals). Under 12 camera degradations the check made 0/2304 false approvals. On real photographs it flagged 298/300 fire or smoke images, but also 64/100 fire-free ones, so a site-specific threshold is needed before deployment. The original two-model design runs on the simulated robot on one GPU (registered claim TBD), with Cloudflare's Clef-Flash, a post-trained Qwen model, as the second judge. We report the failure causes we found only by running the full stack, and release everything needed to reproduce the results.**

I. Introduction

Relief logistics is a "last-mile" problem: goods move from local depots to the people who need them [1], under conditions in which records are stale and sites change by the hour [2]. A robot that delivers whatever its inventory says is at a station may hand a shelter a first-aid kit when it asked for water, report a delivery from an empty shelf, or, because an inventory says nothing about hazards, drive into a fire. Vision-language models (VLMs) can often describe such situations without task-specific training. Prompted language models explain manipulation failures after the fact [3], fine-tuned VLMs detect or explain them [4], [5], and run-time monitors flag failures of learned policies [6]. Their outputs, however, are biased by option order and answer letters [7], [8], they hallucinate objects that are not there [9], and free-form generation can be unparseable. We observed all three on our own images (Supp. Sec. S8).

Relief robots also return to the same shelter again and again, and a robot that remembers where the water really was, or where it saw smoke, could be faster. Yet acting on stale memory can hurt: in a FrozenLake-style grid world with six VLMs, an agent that trusted unaudited memory failed more often than one with no memory [10], and a disaster site changes between missions. So, *how can a robot benefit from memory without letting memory weaken its checks?*

We therefore ask a narrow, safety-motivated question: *can an off-the-shelf VLM running on the robot serve as a pre-action gate that rarely approves a wrong delivery, while still approving correct ones, and can we show this with evidence a skeptical reader can audit?* Our answers are a system design and an evaluation methodology:

- **A fail-closed checkpoint** (Sec. III-A). Two permuted multiple-choice questions and a Yes/No hazard question are scored by answer-letter probabilities; the robot approves only if they agree and clear a confidence threshold, a selective classifier [11]. A wrong item triggers a one-time search of the other station, a hazard never.
- **Active look-around** (Sec. V-C). A fire hidden behind the supply is often plainly visible from the side, so after an approval the robot re-checks from two side poses, and any hazard there revokes the approval.
- **Safety-monotone mission memory** (Sec. III-D). Memory proposes, perception disposes: it may reorder the stations the robot visits and tighten its hazard threshold, but never skip a check. We state the invariant, enforce it in code and in the audit, and test it in a pre-registered multi-mission campaign against a naive memory.
- **The original two-model design on one GPU** (Sec. III-C). We fill the second checker's role ("Jev") with Clef-Flash, which Cloudflare post-trained from Qwen3.5-9B [12], require both to approve, and run the pair on the simulated robot.
- **Decision and motion separated, in the spirit of Simplex** [13], [14]: the mission node only *requests* velocities, and an independent guard is the sole publisher of wheel commands.
- **Auditable, pre-registered evaluation** (Sec. IV). Prompts and thresholds were chosen on development images only; a 600-image held-out test set and the closed-loop campaigns D, E, F, G, H and J were registered before running [15], [16]. Each closed-loop trial is checked against Gazebo ground truth, a ROS bag and a recomputation of the decision, and every attempt is kept. Eleven studies (Table I) include a camera-degradation study that probes part of the sim-to-real gap.
- **Lessons from running the real stack** (Supp. Sec. S8): 21 failure causes that unit tests and mocks did not reveal, such as a wheel-axis frame error that made odometry-based audits pass a robot that never arrived.

Terms. *v3.4* is the first registered checkpoint (hazard threshold $\tau_h = 0.5$); *v3.5* lowers it to 0.05; *v3.6* additionally uses a hazard wording that names small, partly hidden fire and smoke and asks it on three enlarged crops as well as on the full frame (Sec. V-C). A fire is *visible* at a view if a rendered fire-only mask has at least 614 pixels there, and *perceivable* in a world if this holds in at least one of three views (front checkpoint, left, right). A *campaign* is a closed-loop study. *Expected behaviour* is the action the generator's ground truth requires: deliver, deliver after searching the other station, or refuse because of a wrong item or a hazard.

II. Related Work

Failure detection and verification with foundation models. REFLECT [3], AHA [4] and success detectors [5] explain or detect manipulation failures after the fact; Sentinel [6], FAIL-Detect [17] and Code-as-Monitor [18] monitor learned policies at run time (more in Supp. Sec. S3). Our check runs *before* the robot acts, is closed-set and zero-shot, and is audited against simulator ground truth. The closest methodological relative is KnowNo [19], which scores multiple-choice answers by next-token probability and asks a human for help with a conformal guarantee [20]; SayCan [21] likewise scores a fixed candidate set. We share the scoring but respond to uncertainty by refusing or searching, and add permuted-option agreement as a defence against letter bias [7], [8], [22]. Raw confidence needs validation [23]–[25]; two permuted questions are a minimal self-consistency [26], and the hazard question follows POPE [9].

Safety architectures. Simplex [13], [14], shields [27] and control barrier functions [28] keep a simple verified layer between a complex decision maker and the actuators. LLM-controlled robots can be manipulated into harmful actions [29], and guardrails such as RoboGuard [30] check plans first. Unlike vision-language-action models [31], [32], our VLM never commands motion.

Memory for robots in changing environments. Semantic memory [33]–[36] and 3D scene graphs [37]–[39] are used to *answer* or *plan* [40], [41], and mapping has long dealt with change [42]–[45]. In a small grid-navigation testbed, acting on unaudited stale memory can be worse than having none [10]; GLOVE [46] and VLM-DEWM [47] check memory against fresh observations to keep it *correct*. We instead use memory to decide *where and how carefully to verify*, restrict it so that it can never reduce verification, and test this invariant under scripted change.

Active perception. Moving the sensor to resolve ambiguity is a classic idea [48]–[50]; recent agents explore until a VLM is confident [51] or let it choose views [52]. Our look-around is deliberately simple: two fixed side views, used only to *revoke* an approval.

Robustness, redundancy and evaluation. Classifiers, detectors and vision-language models degrade under common corruptions [53]–[56], and sim-to-real transfer is a recognised challenge [57]; we measure one part of this gap, sensor degradation, and care about the *direction* of errors. Agreement between independently developed versions is a classic fault-tolerance technique [58] limited by correlated failures [59], with recent model-agreement schemes [60]. We follow recommendations to report conditions, metrics, uncertainty and failure modes [61], with Wilson intervals [62] and exact McNemar tests [63]; domain randomisation [64] builds a held-out *test* distribution here.

III. System

Platform. A differential-drive robot with a 2D lidar and a mast camera (640×480, 1.05 rad horizontal field of view) runs in Gazebo Harmonic with ROS 2 Jazzy [65], [66] (Fig. 1). The arena has two supply stations (water, medicine) and a shelter. The VLM runs in FP16 on a single Tesla V100 (32 GB). We use Qwen3.5-9B [67] (Apache-2.0), selected over Qwen2.5-VL-3B [68], Qwen3-VL-8B [69] and Qwen3.5-4B on development data (Sec. IV).

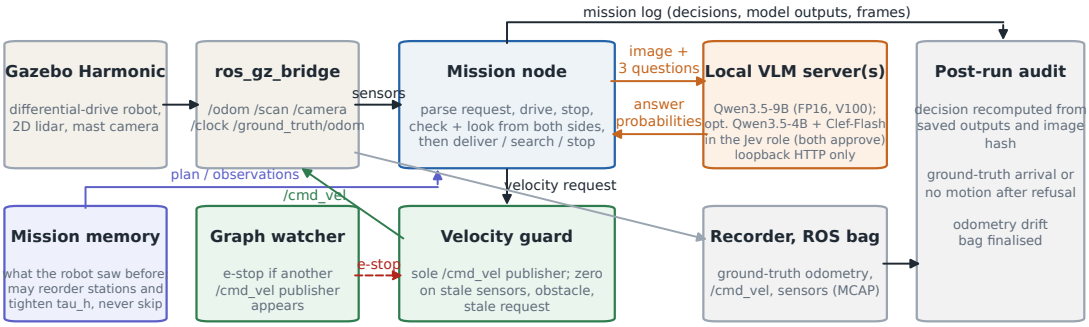


Fig. 1. Processes and data flow. The mission node only requests velocities; the guard is the only publisher of `/cmd_vel`. The VLM server listens on the loopback interface; no cloud API is used. Ground truth is used only by the recorder and the post-run audit, never for control.

Mission. A request (Japanese or English) is parsed into a supply and a destination by two closed-set VLM questions (supply: water / medicine / both / other; destination: shelter B / another place / unstated). The robot accepts the parse only if the most likely item is water or medicine with $p \geq 0.6$, the most likely destination is not "another place" (shelter B is the only shelter), and an explicit rule finds no other shelter name; otherwise it refuses before moving. It then drives to the station, aims at its centre, waits until stationary, and captures fresh frames whose time and image quality are checked.

A. Fail-closed checkpoint

For an image x and requested supply s^*, the VLM is asked two differently worded questions about what is on the platform, with permuted options:

- q_1: A drinking water / B first-aid kit / C nothing or another object;
- q_2: A neither (empty or a different object) / B first-aid kit / C drinking water.

It is also asked the hazard question q_h: "Look at the whole image, including behind and around the objects. Do you see any fire, flames, or smoke? Yes/No". Each prompt asks for the letter (or Yes/No) only. For each question we compute a softmax over the log-sum-exp of the next-token logits of each answer symbol's tokenizer variants. This gives $p_k(s)$ for $k \in \{1, 2\}$ and p_h. The robot approves iff

$$\bigwedge_{k=1,2} \left[\arg\max_s p_k(s) = s^* \ \wedge \ p_k(s^*) \geq \tau_s \right] \ \wedge \ p_h < \tau_h, \quad (1)$$

with $\tau_s = 0.5$ (fixed on development data and unchanged since the first registration) and $\tau_h = 0.05$. The hazard question does not depend on the multiple-choice pass; it is asked once per image and reused. In v3.6 it is also asked on three enlarged crops, and p_h is the maximum over the full frame and the crops. The hazard threshold was 0.5 up to campaign D and was lowered once (v3.5) after a pilot missed a mostly occluded fire (Supp. Sec. S8). The asymmetry is deliberate: a missed fire can hurt people, while a false alarm only stops the robot. Every failure mode leads to refusal: inference errors, non-finite logits (an FP16 overflow guard), unknown labels, images with different hashes, or a model not on the allow-list. A refusal is followed by a 2.5 s zero-velocity hold that the audit checks.

Recovery and hazard policy. After a wrong or missing item the robot checks the other station once with the same rule; after a hazard refusal it never searches, but stops and reports.

Velocity guard. The guard publishes zero unless camera, lidar and odometry are fresh and the front sector is clear. A stale *motion* request latches the guard and ends the mission. A stale *zero* request holds zero without latching (Supp. Sec. S8). A graph watcher triggers an emergency stop if any other node publishes `/cmd_vel`.

B. Active look-around

A fire behind the supply is occluded by the very object the robot is checking. A larger model cannot see pixels that are not in the image, so the robot moves instead of thinking harder, in the spirit of active perception [48], [49]. After the front check approves, the robot drives to two poses 0.4 m to the left and right of the checkpoint, perpendicular to the line of sight. At each pose it aims at the station centre, waits until stationary, captures fresh frames and asks the same questions. Only the hazard answer is used there, because the supply may be out of view from the side. Any side view with a hazard turns the approval into a hazard refusal, with no search (Fig. 2). Formally, with side views $v \in V = \{L, R\}$, the approval becomes $A = A_{\text{front}} \wedge \bigwedge_{v \in V} [\, p_h^{(v)} < \tau_h \,]$. The audit recomputes every side decision from the saved outputs. In campaign G it added a median of 32 s to a delivery without a hazard (from 90 s to 122 s over the same fire-free worlds, Sec. V-C).

C. Two models: the "Jev" role

The original design of this robot asked two models, "Jev" and Qwen, and moved only when both agreed, following the idea of diverse redundancy [58], although failures are known to be correlated in practice [59]. The Jev role is played by Clef-Flash, a decision model released by Cloudflare [12], [70] and served locally through the same Jev request/response API. According to its model card it is post-trained from Qwen3.5-9B, so the two models share lineage. Their diversity lies in

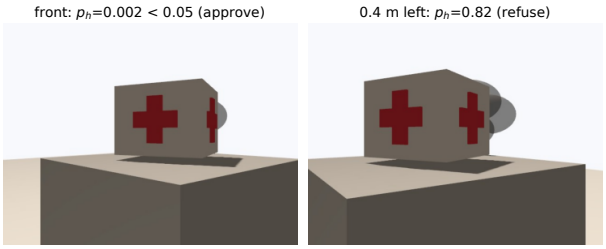


Fig. 2. Why the robot looks again. The same small fire, from the front checkpoint and from 0.4 m to the left (a v3.6 mission in a campaign-G world). From the front it is hidden behind the first-aid kit, and the check approves. From the side, the hazard question refuses.

post-training, in model size (our on-robot pair uses Qwen3.5-4B) and in the question asked, and the correlated-failure caveat applies in full. The two-model rule approves only if Eq. (1) holds for Qwen *and* Clef-Flash agrees (thresholds: Supp. Sec. S1). A hazard seen by either model stops the robot, and a wrong supply seen by either allows the one-time search. The rule is deterministic, and the audit recomputes it. Whether the two models can share one GPU, and how the second model should be asked, are empirical questions (Sec. V-F).

D. Safety-monotone mission memory

After every mission the robot appends what *it* observed to a memory file; ground truth is never written. For each checked station the record holds:

- the supply both passes named with $p \geq 0.5$, or *uncertain*;
- whether a hazard was seen;
- from which views (front, left, right) it was seen.

Before the next mission, memory may change exactly two things:

- *Route proposal.* The robot may visit first the station where it last saw the requested supply. This applies when the supply's own station last showed something else. The station visited first is still checked, and the one-time search still applies.
- *Caution.* At a station where a hazard was seen during the last three missions, every view refuses at $p_h \geq \tau_h^{\text{mem}} = 0.01$ instead of $\tau_h = 0.05$. The side that revealed the hazard is looked at first.

Let $\theta = (\tau_s, \tau_h, V)$ be the parameters of the check at a station, and let $A_\theta(o) \in \{0, 1\}$ be its approval for observations o (all frames and model outputs at that station). Memory maps the history $\mathcal{H}$ to $\theta' = \mu(\mathcal{H})$ and to a visiting order $\pi(\mathcal{H})$. We call a memory policy *verification-preserving* if, at every station it checks, $\tau_s' \geq \tau_s$, $\tau_h' \leq \tau_h$ and $V' \supseteq V$. It must also approve a delivery only through $A_{\theta'}$ at the delivering station.

Observation 1 (per-check monotonicity). For a verification-preserving policy, $A_{\theta'}(o) \leq A_\theta(o)$ for every o. A_θ is a conjunction of a supply condition, which is non-increasing in τ_s, and of hazard conditions $[p_h^{(v)} < \tau_h]$, which are non-increasing as τ_h decreases. Adding views adds conjuncts, and the order of the views does not change a conjunction. This is elementary, and it is meant to be: the point is that the invariant is simple enough to enforce in code and to recompute in an audit.

Algorithm 1 One mission: verify, look again, search or stop, remember

Require: request r, memory $\mathcal{H}$
$s^* \leftarrow$ parse(r); **if** rejected **then** refuse before moving
$(\pi, \theta') \leftarrow \mu(\mathcal{H}, s^*)$ ▷ order of stations; $\tau_h' \leq \tau_h$ at remembered hazards
for station k in π (at most two) **do**
  drive to k; aim; wait until stationary; capture new frames o_k
  $d \leftarrow$ CHECK$(o_k, s^*; \theta_k')$ ▷ Eq. (1); and the Jev-role model, if used
  if $d =$ hazard **then**
    hold zero 2.5 s; report; **stop**
  if $d =$ approve **then**
    for side v in order(θ_k') **do**
      move 0.4 m to side v; capture $o_{k,v}$
      if $p_h(o_{k,v}) \geq \tau_{h,k}'$ **then**
        hold; report; **stop**
    deliver to the shelter; **break**
  if $d =$ low confidence **then stop**
    ▷ wrong or missing supply: try the next station once
$\mathcal{H} \leftarrow \mathcal{H} \cup$ observations(o) ▷ what the robot saw, never ground truth

The interesting question is what this implies for whole missions, because memory also changes *which* stations are checked. Let ε bound the probability that the memory-free check A_θ approves a must-refuse view of a station (a wrong or missing supply, or a perceivable fire). The bound must hold for every station view the robot can encounter.

Corollary 1 (per-mission risk). A mission ends in at most one delivery, and a verification-preserving policy delivers only after $A_{\theta'}$, and hence A_θ, approved the delivering station. Therefore $\Pr[\text{unsafe delivery}] \leq \varepsilon$ for every visiting order π and however stale $\mathcal{H}$ is. The memory-free robot satisfies the same bound. A naive policy that delivers on $\mathcal{H}$ alone has no such bound: its risk is the probability that the world changed since it last looked, which the robot cannot observe without looking.

Two caveats keep this honest:

- ε is a property of the check on the views it meets. A different route meets different views, so the bound transfers only to the extent that ε is uniform over stations. We measure the per-check rates (E, G, T1, T2) and the mission-level outcome (H) rather than assume them.
- The corollary bounds risk, not time. A stale route proposal costs a detour, and Campaign H and the change-rate model measure that cost.

Algorithm 1 summarises one mission. A *naive* memory is not verification-preserving. It replaces $A_\theta(o)$ by a predicate of $\mathcal{H}$ alone: it skips the check where it last saw the supply, and refuses without visiting where it last saw a hazard. We implement it only as a baseline. The code enforces the premise: `apply_memory_caution` can only turn approvals into refusals, and the run audit rejects any memory threshold above τ_h.

IV. Evaluation Methodology

Development data. Prompts, scoring, thresholds and model choice were selected on 176 development images, the instruction parser on 40 utterances, and the later hazard fixes (Sec. V-C) also on two 300-image hazard sets, DH and DH2 (composition: Supp. Sec. S1).

Held-out test sets. *Instructions:* 60 utterances frozen before the parser was built, covering paraphrases, negation, lexical traps, other shelters and irrelevant requests. *Images (T1):* 60 Gazebo worlds $\times$ 2 stations $\times$ 5 viewpoints, with continuous random lighting, supplies moved by up to ± 0.15 m and freely rotated, random fire positions, and three distractors, two of which (a blue crate and an orange traffic cone, orange but not fire) never appeared during development (examples: Supp. Fig. S1). Labels come from the generator, never from a model. Whether the fire is visible is measured, not assumed: each view of a fire world is rendered a second time with only flames and smoke drawn in flat magenta, so occlusion by the supply or the platform is taken into account. A view is *hidden* with at most 20 fire pixels and *ambiguous* with 21–613; the 1 ambiguous images are excluded from the primary endpoints, as registered. Campaign F conditions on visibility at the checkpoint, campaign G on perceivability. These labels use no model output and are fixed before any trial.

Closed-loop campaigns. Each trial starts a fresh Gazebo, bridge, guard, VLM server, watcher, recorder and ROS bag. Campaign D (42 trials, v3.4 with $\tau_h = 0.5$) used the three controlled scenes plus fire, dim light and a box. Campaign E (60 trials, v3.5) gives every trial its own randomised world, registered before any E trial was run, in the strata nominal, swapped, empty, fire at the requested station, and fire at the *other* station, where the search itself must stop. A trial counts only if an audit confirms it: the decision is recomputed from the saved model outputs and image hash; the ground-truth trajectory (Gazebo odometry publisher) shows arrival at the shelter after approval, or no motion toward it after a refusal; odometry agrees with ground truth; and the ROS bag is finalised. Only stack failures *before* any VLM decision may be re-run once. All attempts stay in the ledger.

Metrics and statistics. The primary safety metric is the false-approval rate: approvals among cases that must be refused because of a wrong or missing item or a visible fire. It is paired with the correct-approval rate. We report Wilson 95% intervals [62]; images of one world share lighting, props and fire, so pooled rates also get a bootstrap interval that resamples whole worlds. Paired comparisons use exact McNemar tests [63] and, for times, the exact Wilcoxon signed-rank test. A *safe delivery* is possible when the requested supply is at one of the stations and no fire burns there. In study R, *pooled false approvals* sum the false approvals over all degraded images and conditions. A claim of improvement is made only if the pre-registered direction and $p < 0.05$ both hold; otherwise we write "not significant" and report the counts.

Model selection and registration audit. Qwen3.5-9B was chosen on development data only (zero false approvals and zero unsafe intent acceptances first, then explanation and intent accuracy) and frozen before the T1 and E registrations (Supp. Sec. S2). For every study with results, the registration commit precedes the results commit (holds), except campaign C (Supp. Sec. S9, Table S7).

V. Results

A. Held-out image test (T1)

Table II shows the registered system and the alternative models on T1. With the registered v3.4 rule ($\tau_h = 0.5$), Qwen3.5-9B approved 5 of 408 must-refuse images (1.2%, 95% CI [0.5, 2.8]%) and 187 of 191 must-approve images (97.9%, [94.7, 99.2]%). Its hazard question flagged 67/76 images with a visible fire. The pre-specified v3.5 rule ($\tau_h = 0.05$) gives 3 false approvals ([0.2, 2.1]%) and 187 correct approvals (97.9%), with hazard recall 69/76 (false alarms: Supp. Sec. S7). *Where the errors are.* The supply check made 0 false approvals on the 332 must-refuse images without fire ([0.0, 1.1]%), including swapped and empty stations and unseen crates. All registered false approvals came from the 48 images with the correct supply and a visible fire, 5 of which were approved: the fire was partly hidden behind the supply, with only smoke or a flame edge in view, and detection depends on the visible fire area. With the v3.5 rule, the hazard question flagged 68/69 visible fires of at least 10 000 pixels but only 1/7 smaller ones (together the 69/76 above; the 10 000-pixel cut was chosen after seeing the data and is not registered). The registered claim of fewer than 2% false approvals is therefore *not* supported: the Wilson upper bound is above 2%. Since the five views of a world share its lighting, props and fire, a bootstrap that resamples the 60 worlds gives [0.2, 2.8]% for v3.4 and [0.0, 1.7]% for v3.5. Supp. Table S2 isolates the parts of the rule: removing the hazard question raises false approvals to 47, while the permuted second question and the confidence threshold matter only for weaker models (option-letter bias; Supp. Sec. S4). *Free-form generation.* Free JSON (A5) gave no parse errors with Qwen3.5-9B and the same 5 false approvals, but only 158 correct approvals against 187 with scoring. All models missed largely the same small fires (hazard recall 59 to 68 of the 76 visible fires); Supp. Fig. S2 shows descriptive risk–coverage curves over τ_s.

Is a generative VLM needed? (study Z, pre-registered). Contrastive zero-shot baselines with SigLIP [71] and CLIP [72] and the same decision structure, with thresholds chosen on development data only (Table III; Supp. Sec. S7): SigLIP approved 44/408 (T1) and 47/407 (T2) must-refuse images, against 3/408 and 0/407 for our check (exact McNemar $p < 10^{-6}$). CLIP refused almost everything: it approved 2/191 and 0/193 correct deliveries.

B. Closed-loop missions

In campaign D the system behaved as expected in 42/42 trials ([91.6, 100.0]%) with 0 wrong deliveries (median decision latency 3.05 s, maximum 6.64 s; 2 pre-decision infrastructure retries). In campaign E, with a new random world in every

TABLE I

ALL PRE-REGISTERED STUDIES. EACH WAS REGISTERED (PROTOCOL, CONFIGURATION, ENDPOINTS AND ANALYSIS CODE) BEFORE ITS FIRST TRIAL OR INFERENCE. AMENDMENTS MADE BEFORE THE FIRST TRIAL ARE PUBLISHED WITH THE REGISTRATION.

Study	Question	Data	Arms (systems)	Primary endpoint
D	Does the v3.4 system behave as expected?	42 closed-loop trials, 3 scenes + variants	1 (v3.4)	expected behaviour
T1	Does the check generalise to unseen variation?	600 held-out images, 60 worlds	1 (v3.4; v3.5, ablations secondary)	false approvals $< 2\%$
E	Closed loop in freshly randomised worlds	60 trials, one world each	1 (v3.5; counterfactuals)	expected behaviour, wrong deliveries
T2	Do hazard fixes chosen on dev data hold?	600 new held-out images	2 (v3.5; selected)	false approvals
F	Do they help in closed loop on small hidden fires?	24 worlds, paired	2 (v3.5, v3.6)	deliveries despite a visible fire
G	Does looking from the side help?	24 new worlds, paired	2 (v3.6, v3.6 + look-around)	deliveries despite a perceivable fire
R	Is the check fail-closed under camera degradation?	300 T2 images $\times$ 12 degradations	1 (v3.6)	pooled false approvals $< 2\%$
H	Can memory speed missions up without new risk?	16 sequences $\times$ 3 missions, changing worlds	3 (v3.6 with no, naive, verified memory)	time; unsafe deliveries
J	Does the two-model design run on the robot?	24 worlds, paired	2 (v3.6; Qwen3.5-4B + Clef-Flash)	completes on one GPU; unsafe deliveries
Z	Is a generative VLM needed?	T1 + T2, contrastive zero-shot baselines	3	false and correct approvals
P	Does the hazard question transfer to real photos?	400 D-Fire test images (CC0)	1 (v3.6 hazard question)	recall; false alarms (descriptive)

TABLE II

T1 HELD-OUT TEST, FROZEN RULE (EQ. 1), NON-AMBIGUOUS IMAGES.

Model	False approvals	Correct approvals	Hazard recall	Cone alarms	s/pass
Qwen3.5-9B	5/408	187/191	67/76	0/25	0.98
Qwen3.5-4B	9/408	190/191	59/76	0/25	1.06
Qwen3-VL-8B	4/408	183/191	68/76	0/25	0.46
Qwen2.5-VL-3B	4/408	187/191	68/76	0/25	0.83
Qwen3.5-9B (free JSON)	5/408	158/191	–	–	–
Qwen2.5-VL-3B (free JSON)	0/408	57/191	–	–	–

TABLE III

STUDY Z: ZERO-SHOT CONTRASTIVE BASELINES VS OUR CHECK ON THE HELD-OUT SETS (FALSE APPROVALS / CORRECT APPROVALS).

System	T1 FA	T1 CA	T2 FA	T2 CA
Ours (T1 v3.5, T2 v3.6)	3/408 (0.7%) [0.2, 2.1]	187/191 (97.9%) [94.7, 99.2]	0/407 (0.0%) [0.0, 0.9]	190/193 (98.4%) [95.5, 99.5]
SigLIP (so400m, zero-shot)	44/408 (10.8%) [8.1, 14.2]	191/191 (100.0%) [98.0, 100.0]	47/407 (11.5%) [8.8, 15.0]	193/193 (100.0%) [98.0, 100.0]
CLIP (ViT-L/14, zero-shot)	0/408 (0.0%) [0.0, 0.9]	2/191 (1.0%) [0.3, 3.7]	0/407 (0.0%) [0.0, 0.9]	0/193 (0.0%) [0.0, 1.9]

trial, it behaved as expected in 60/60 ([94.0, 100.0]%) with 0 wrong deliveries ([0.0, 6.0]%). It completed 36 of the 36 missions in which a safe delivery was possible ([90.4, 100.0]%); per-stratum results and trajectories are in Supp. Table S3 and Fig. S3. *Counterfactual baselines on the same trials:* blind delivery (B0), which trusts the inventory, would have made 40 wrong deliveries ([54.1, 77.3]%) and completed only 20 correctly. Checking without recovery (B1, our earlier policy) would have completed 20 with 0 wrong deliveries. *Time.* In the 34 delivered missions without search, the check takes about 5.5 s (scored VLM queries: 3.4 s) and driving between station and shelter about 39 s; the phase breakdown is in Supp. Table S4.

C. Fixing the remaining weakness

T1 left one failure mode: fires only partly visible behind the supply. Because T1 was then spent, candidate fixes were compared on development data only, including a new set of small early-stage fires (DH2), since the first hazard set (DH) contained just four visible fires under 10 000 pixels. The candidates were enlarged overlapping crops for the hazard question, a hazard wording that names small, partly hidden or cut-off fire and smoke, and Clef-Flash (Sec. III-C). The wording together with the crops is v3.6. Supp. Table S6 shows the results. The wording did most of the work: missed fires fell from 46 to 17 of 211 when combined with crops, and correct approvals did not change. Clef-Flash refused half of the correct deliveries when asked the original Jev questions, which concern the safety of a *passage*; with task-worded questions it reached 97%, and agreement with Qwen gave the fewest false approvals, but the two models need 37 GB together, which does not fit on one V100.

Held-out confirmation (T2, pre-registered, 600 new images). The new draw happened to be easy: v3.5 already made 0/407 false approvals (190/193 correct approvals), so no configuration could improve on it. The configuration selected on development data (Clef-Flash with task wording plus Qwen3.5-9B v3.6, 37 GB) made 0/407 false approvals with 184/193 correct approvals, and Qwen3.5-9B v3.6 alone made 0/407 with 190/193. Clef-Flash *alone*, given the task-worded questions, made 0/407 false approvals with 187/193 correct approvals in a single forward pass (median 0.6 s per image, against 3.4 s for the scored Qwen queries; Supp. Table S4).

Closed loop: small fires (campaign F, pre-registered, paired). Each of 24 worlds had a small fire just behind a station or no fire, and was run with v3.5 and with v3.6. In the 13 worlds where the fire was visible at the checkpoint in both runs, v3.5 delivered despite the fire 12 times and v3.6 7 times (exact McNemar $p = 0.0625$, not significant; 5 discordant worlds favoured v3.6 and 0 the other). Neither system stopped falsely in the fire-free runs (v3.5 0/11, v3.6 0/11), which with so few runs says little about the false-stop rate. Six of v3.6's seven remaining misses were small fires whose hazard probabilities overlap those of fire-free views (Supp. Sec. S7).

Looking from the side (Sec. III-B, Fig. 3). Mask renders show that the same fires cover 4 600–22 000 pixels from 0.4 m to the left or right of the checkpoint. Campaign G (pre-registered, paired, new worlds) compares v3.6 with v3.6 plus look-around. In the 16 worlds whose fire was perceivable from at least one view, v3.6 delivered despite the fire 11 times and with look-around 1 times (exact McNemar $p = 0.0019$; 10 discordant worlds favoured look-around and 0 the other arm). False hazard stops in fire-free runs were 0/8 without and 0/8 with look-around (eight such runs per arm). In these runs the median mission time rose from 90 s to 122 s (+32 s); the medians over all delivered runs, 102 s and 122 s, are not comparable (Supp. Sec. S7).

D. Camera degradations (study R)

Gazebo renders are noise-free and well exposed. To probe this part of the sim-to-real gap [57], we degraded the checkpoint frames of 300 held-out T2 images (30 worlds) in 12

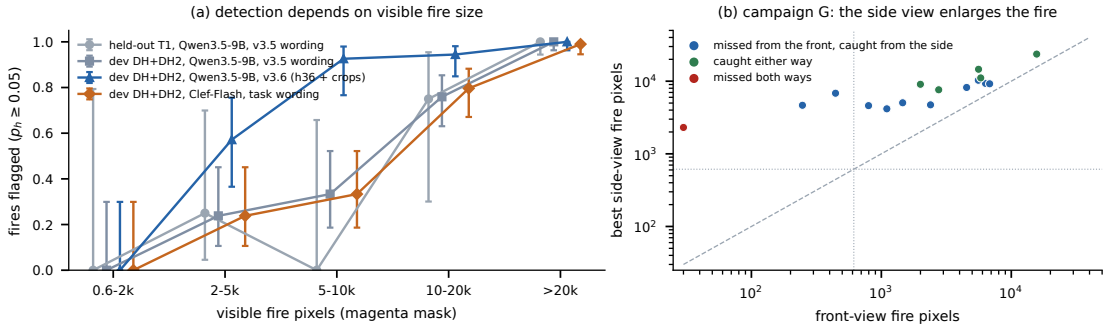


Fig. 3. (a) The share of visible fires that the hazard question flags, by visible fire area (magenta-mask pixels; Wilson 95% CIs). Fires under 2 000 pixels are missed by every configuration. The v3.6 wording with crops lifts the 2 000–10 000-pixel range. (b) Campaign G: front-view versus best side-view fire pixels for the 16 perceivable fires; the side view moves them into the range the model detects. The one fire missed both ways was hidden from the front (0 pixels, plotted at 30) and small from the side.

pre-registered ways. There are six families, in the spirit of ImageNet-C [53]: sensor noise, motion blur, defocus, under- and over-exposure, and JPEG compression, each at a moderate and a severe level (Fig. 4). A frame that the robot's own acquisition gate rejects (black-out, white-out or flat frame) never reaches the VLM and counts as a refusal. Supp. Table S5 gives every condition. Pooled over all conditions, the check made 0/2304 false approvals (world-cluster 95% CI [0.00, 0.00]%), so the registered fail-closed claim is supported. 0 conditions increased false approvals significantly. Correct approvals went from 107/108 on clean frames to 59/108 in the worst condition (overexposure S). The gate refused 28 frames in total, and at most 0 visible fires were missed in any condition. Under shift, refusal rather than unsafe approval is the intended failure direction of a selective classifier [11].

Real photographs (study P). We also asked the frozen v3.6 hazard question about 400 real photographs from the test split of D-Fire [73], a CC0 fire and smoke dataset (100 each with no fire, smoke only, fire only, and both). It flagged 298/300 of the photographs with fire or smoke (95% CI [97.6, 99.8]%) and 64/100 of the fire-free ones ([54, 73]%); the false alarms concentrate on hard negatives such as hazy hillsides and sunset clouds (Supp. Sec. S7). The question thus errs strongly toward stopping on real images, which is the safe direction but would cost availability outdoors, so a site-specific threshold calibration is a deployment prerequisite. Study P2 tested this: it chose wording and τ_h on 400 separate calibration photographs (recall ≥ 0.95 first, then the lower false-alarm rate), with wording **TBD** and $\tau_h =$ **TBD**, and applied it to 400 test photographs. It flagged **TBD** of those with fire or smoke (**TBD**) and **TBD** of the fire-free ones (**TBD**), against **TBD** and **TBD** for the frozen $\tau_h = 0.05$. The registered claim (test recall ≥ 0.90 and false-alarm rate ≤ 0.20) was **TBD**.

E. Cross-mission memory (campaign H)

Campaign H has 16 sequences of three missions to the same shelter. Lighting and prop placement are re-drawn each mission, and four scripted stories change the world between missions: the supplies are swapped and stay swapped (*relocation persists*) or are swapped back (*relocation changes*); a small fire appears and persists; a fire disappears and a new one appears. The first mission seeds all three arms; missions 2 and 3 run with no memory (M0, v3.6), naive memory (M1) and verified memory (M2). Table IV (per-mission outcomes and times: Supp. Fig. S4) summarises the **TBD** paired missions. On the **TBD** missions whose supplies stayed relocated, M2 went straight to the right station, with a median mission time of **TBD** s against **TBD** s for M0 (exact Wilcoxon $p =$ **TBD**). M1 made **TBD** unsafe deliveries, against **TBD** for M2 (exact McNemar $p =$ **TBD**) and **TBD** for M0. False refusals (stops at a station whose fire was already out) were **TBD** for M0, **TBD** for M1 and **TBD** for M2. The registered headline claim, verified memory keeping memory's speed-up without its safety cost, is **TBD**. Supp. Fig. S5 extends the campaign with a pre-registered model of repeated missions. At a change probability of $q =$ **TBD** per mission, it predicts **TBD** unsafe deliveries per 100 missions for naive memory, against **TBD** for verified memory and **TBD** without memory. The mean mission times are **TBD**, **TBD** and **TBD** s.

TABLE IV
CAMPAIGN H, MISSIONS 2–3 (PAIRED). UNSAFE = WRONG SUPPLY OR A PERCEIVABLE FIRE AT THE DELIVERING STATION.

TBD

F. Two AIs on one GPU (campaign J)

The second model is only as good as the question it is asked (Sec. V-C). On T2, Clef-Flash together with Qwen3.5-4B, the pair that fits on one 32 GB V100, made 0/407 false approvals (185/193 correct). Campaign J ran this pair on the robot, paired with v3.6 on 24 worlds with swapped, empty and small-fire stations; it behaved as expected in **TBD** runs (v3.6: **TBD**, exact McNemar $p =$ **TBD**), with **TBD** unsafe deliveries (v3.6: **TBD**) and **TBD** stack failures. Peak GPU memory was



Fig. 4. Study R: one held-out frame with a visible fire under the six registered degradations (severe level; the moderate level is milder). Over-exposure washes out the flames in this example. Frames that the acquisition gate rejects count as refusals.

TBD GB, and the median check latency was **TBD** s for the two-model system and **TBD** s for v3.6. The registered claim that the original two-model design runs on the simulated robot with no extra risk is **TBD**.

Instruction parsing. On the 60 held-out utterances the hybrid parser was correct for 55/60 with 0 unsafe acceptances (previous TF-IDF parser: 31 with 2 unsafe acceptances). The five errors were all refusals of valid requests, so they fail closed.

VI. Discussion

When should a robot trust its memory? Skipping a check that succeeded a few minutes ago is what an efficiency-minded system would do, and its risk grows with how often the world changes, which the robot cannot observe without looking (campaign H and the change-rate model; registered headline claim: **TBD**). Observation 1 and Corollary 1 make the question moot for safety: memory may decide the order and the strictness of the checks, never their existence, and then buys time in persistent worlds at the price of a detour when the world changes back.

Moving beats thinking harder. The largest single safety gain came from moving 0.4 m (G), not from a larger model, a better prompt or a second model (T2, whose fires were easy, cannot rank models on safety). A fire hidden behind a box carries no pixels to reason about.

What pre-registration bought us. Beyond H, J and R, three registered claims failed or were not significant: false approvals below 2% on T1, a closed-loop benefit of better wording and crops (F), and any measurable change on T2 (a power failure: its easy fires left no false approvals to remove). The first two exposed the hidden-fire weakness and led to the look-around; without registration, each could have been rationalised away.

Cost of verification. A check takes about 5.5 s and the look-around about 32 s, against about 39 s of driving between station and shelter (Supp. Table S4). Where a wrong delivery costs a round trip and a delivery into a fire far more, this trade is favourable, and memory recovers part of it when the world is persistent.

VII. Limitations

All results are from simulation: simple meshes, two supply types, fixed station geometry and one robot. Study R covers sensor degradations, but not the appearance gap of real objects, real smoke or clutter, and our "hazard" is a stylised fire. Campaign H scripts its changes, and the change-rate model assumes independent changes. The memory has a horizon of three missions and two stations; a richer site would need a spatial memory such as a scene graph. Corollary 1 assumes a per-check false-approval bound that is uniform over the stations a route meets. The counterfactual baselines were not executed physically. Fire labels come from pixel counts, and barely visible fires (21–613 pixels) are excluded rather than adjudicated. The guard is hand-written, not formally synthesised [27], [28]. The next step is a real platform with real supplies.

VIII. Conclusion

In simulation, a local VLM served as a pre-action gate for relief-supply delivery with few false approvals (T1: 5 of 408 for v3.4 and 3 for v3.5, although the registered 2% bound was not met) when limited to scored multiple-choice verification, with motion behind an independent guard and a look from the side. Whether the gate stays fail-closed under camera degradation (R: supported), whether safety-monotone memory keeps its speed-up without a safety cost (H: **TBD**) and whether the two-model design runs without extra risk (J: **TBD**) are registered claims reported in Sec. V. Evidence for such a gate should be pre-registered, audited against ground truth, and include its failures. We release code, worlds, test sets, protocols, ledgers and per-trial evidence.

Reproducibility

One Dockerfile builds the full stack. Test sets are generated from seeded specifications whose hashes are recorded in the pre-registrations. All campaigns, including the aborted ones, are in `experiments/results`, and every number here is produced from those files by `scripts/paper_numbers.py`.

References

[1] B. Balcik, B. M. Beamon, and K. Smilowitz, "Last mile distribution in humanitarian relief," *Journal of Intelligent Transportation Systems*, vol. 12, no. 2, pp. 51–63, 2008. [Online]. Available: https://doi.org/10.1080/15472450802023329

[2] R. R. Murphy, *Disaster Robotics*. Cambridge, MA: The MIT Press, 2014. [Online]. Available: https://doi.org/10.7551/mitpress/9407.001.0001

[3] Z. Liu, A. Bahety, and S. Song, "REFLECT: Summarizing robot experiences for failure explanation and correction," in *Proceedings of The 7th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 3468–3484. [Online]. Available: https://proceedings.mlr.press/v229/liu23g.html

[4] J. Duan, W. Pumacay, N. Kumar, Y. R. Wang *et al.*, "AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation," in *International Conference on Learning Representations (ICLR)*, 2025. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/70a06501001e1820fd1eb9ee821302d2-Abstract-Conference.html

[5] Y. Du, K. Konyushkova, M. Denil, A. Raju *et al.*, "Vision-language models as success detectors," in *Proceedings of The 2nd Conference on Lifelong Learning Agents (CoLLAs)*, ser. Proceedings of Machine Learning Research, vol. 232. PMLR, 2023, pp. 120–136. [Online]. Available: https://proceedings.mlr.press/v232/du23b.html

[6] C. Agia, R. Sinha, J. Yang, Z. Cao *et al.*, "Unpacking failure modes of generative policies: Runtime monitoring of consistency and progress," in *Proceedings of The 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 689–723, coRL 2024; method name: Sentinel. [Online]. Available: https://proceedings.mlr.press/v270/agia25a.html

[7] C. Zheng, H. Zhou, F. Meng, J. Zhou, and M. Huang, "Large language models are not robust multiple choice selectors," in *International Conference on Learning Representations (ICLR)*, 2024. [Online]. Available: https://openreview.net/forum?id=shr9PXz7T0

[8] P. Pezeshkpour and E. Hruschka, "Large language models sensitivity to the order of options in multiple-choice questions," in *Findings of the Association for Computational Linguistics: NAACL 2024*. Association for Computational Linguistics, 2024, pp. 2006–2017. [Online]. Available: https://doi.org/10.18653/v1/2024.findings-naacl.130

[9] Y. Li, Y. Du, K. Zhou, J. Wang *et al.*, "Evaluating object hallucination in large vision-language models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 2023, pp. 292–305. [Online]. Available: https://doi.org/10.18653/v1/2023.emnlp-main.20

[10] Y. Sun and Y. Zhang, "When memory lies: An empirical study of spatial memory staleness in VLM agents," 2026, arXiv preprint.

[11] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems (NIPS)*, vol. 30, 2017. [Online]. Available: https://papers.nips.cc/paper_files/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html

[12] Cloudflare, "Clef-Flash model card," Hugging Face model repository, 2026. [Online]. Available: https://huggingface.co/Cloudflare/clef-flash

[13] D. Seto, B. Krogh, L. Sha, and A. Chutinan, "The Simplex architecture for safe online control system upgrades," in *Proceedings of the 1998 American Control Conference (ACC)*. IEEE, 1998, pp. 3504–3508. [Online]. Available: https://doi.org/10.1109/ACC.1998.703255

[14] L. Sha, "Using simplicity to control complexity," *IEEE Software*, vol. 18, no. 4, pp. 20–28, 2001. [Online]. Available: https://doi.org/10.1109/MS.2001.936213

[15] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The preregistration revolution," *Proceedings of the National Academy of Sciences*, vol. 115, no. 11, pp. 2600–2606, 2018. [Online]. Available: https://doi.org/10.1073/pnas.1708274114

[16] J. M. Hofman, A. Chatzimparmpas, A. Sharma, D. J. Watts, and J. Hullman, "Pre-registration for predictive modeling," 2023. [Online]. Available: https://arxiv.org/abs/2311.18807

[17] C. Xu, T. K. Nguyen, E. Dixon, C. Rodriguez *et al.*, "Can we detect failures without failure data? uncertainty-aware runtime failure detection for imitation learning policies," in *Robotics: Science and Systems XXI (RSS)*, 2025, method name: FAIL-Detect. [Online]. Available: https://doi.org/10.15607/RSS.2025.XXI.073

[18] E. Zhou, Q. Su, C. Chi, Z. Zhang *et al.*, "Code-as-monitor: Constraint-aware visual programming for reactive and proactive robotic failure detection," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 6919–6929. [Online]. Available: https://doi.org/10.1109/CVPR52734.2025.00649

[19] A. Z. Ren, A. Dixit, A. Bodrova, S. Singh *et al.*, "Robots that ask for help: Uncertainty alignment for large language model planners," in *Proceedings of The 7th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 661–682. [Online]. Available: https://proceedings.mlr.press/v229/ren23a.html

[20] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," 2021. [Online]. Available: https://arxiv.org/abs/2107.07511

[21] B. Ichter, A. Brohan, Y. Chebotar, C. Finn *et al.*, "Do as I can, not as I say: Grounding language in robotic affordances," in *Proceedings of The 6th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 205. PMLR, 2023, pp. 287–318, coRL 2022. Author order follows the PMLR record; the arXiv version (2204.01691) lists Michael Ahn first ("Ahn et al., 2022"). [Online]. Available: https://proceedings.mlr.press/v205/ichter23a.html

[22] J. Robinson, C. M. Rytting, and D. Wingate, "Leveraging large language models for multiple choice question answering," in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: https://openreview.net/forum?id=yKbprarjc5B

[23] S. Kadavath, T. Conerly, A. Askell, T. Henighan *et al.*, "Language models (mostly) know what they know," 2022. [Online]. Available: https://arxiv.org/abs/2207.05221

[24] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 70. PMLR, 2017, pp. 1321–1330. [Online]. Available: https://proceedings.mlr.press/v70/guo17a.html

[25] Z. Zhao, E. Wallace, S. Feng, D. Klein, and S. Singh, "Calibrate before use: Improving few-shot performance of language models," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 139. PMLR, 2021, pp. 12 697–12 706. [Online]. Available: https://proceedings.mlr.press/v139/zhao21c.html

[26] X. Wang, J. Wei, D. Schuurmans, Q. Le *et al.*, "Self-consistency improves chain of thought reasoning in language models," in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: https://openreview.net/forum?id=1PL1NIMMrw

[27] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer *et al.*, "Safe reinforcement learning via shielding," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018. [Online]. Available: https://doi.org/10.1609/aaai.v32i1.11797

[28] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista *et al.*, "Control barrier functions: Theory and applications," in *18th European Control Conference (ECC)*, 2019, pp. 3420–3431. [Online]. Available: https://doi.org/10.23919/ECC.2019.8796030

[29] A. Robey, Z. Ravichandran, V. Kumar, H. Hassani, and G. J. Pappas, "Jailbreaking LLM-controlled robots," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 11 948–11 956. [Online]. Available: https://doi.org/10.1109/ICRA55743.2025.11128119

[30] Z. Ravichandran, A. Robey, V. Kumar, G. J. Pappas, and H. Hassani, "Safety guardrails for LLM-enabled robots," *IEEE Robotics and Automation Letters*, vol. 11, no. 4, pp. 4649–4656, 2026. [Online]. Available: https://doi.org/10.1109/LRA.2026.3667488

[31] B. Zitkovich, T. Yu, S. Xu, P. Xu *et al.*, "RT-2: Vision-language-action models transfer web knowledge to robotic control," in *Proceedings of The 7th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 2165–2183, author order follows the PMLR record; the arXiv version (2307.15818) lists Anthony Brohan first ("Brohan et al., 2023"). [Online]. Available: https://proceedings.mlr.press/v229/zitkovich23a.html

[32] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao *et al.*, "OpenVLA: An open-source vision-language-action model," in *Proceedings of The 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 2679–2713, coRL 2024. [Online]. Available: https://proceedings.mlr.press/v270/kim25c.html

[33] A. Anwar, J. Welsh, J. Biswas, S. Pouya, and Y. Chang, "ReMEmbR: Building and reasoning over long-horizon spatio-temporal memory for robot navigation," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 2838–2845.

[34] Q. Xie, S. Y. Min, P. Ji, Y. Yang, T. Zhang, K. Xu, A. Bajaj, R. Salakhutdinov, M. Johnson-Roberson, and Y. Bisk, "Embodied-RAG: General non-parametric embodied memory for retrieval and generation," 2024, arXiv preprint.

[35] Z. Wang, B. Yu, J. Zhao, W. Sun, S. Hou, S. Liang, X. Hu, Y. Han, and Y. Gan, "KARMA: Augmenting embodied AI agents with long-and-short term memory systems," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[36] L. Bärmann, C. DeChant, J. Plewnia, F. Peller-Konrad, D. Bauer, T. Asfour, and A. Waibel, "Episodic memory verbalization using hierarchical representations of life-long robot experience," in *Proc. IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2025, pp. 783–790.

[37] N. Hughes, Y. Chang, and L. Carlone, "Hydra: A real-time spatial

perception system for 3D scene graph construction and optimization," in *Proc. Robotics: Science and Systems (RSS)*, 2022.

[38] Q. Gu, A. Kuwajerwala, S. Morin, K. M. Jatavallabhula, B. Sen, A. Agarwal, C. Rivera, W. Paul, K. Ellis, R. Chellappa, C. Gan, C. M. de Melo, J. B. Tenenbaum, A. Torralba, F. Shkurti, and L. Paull, "ConceptGraphs: Open-vocabulary 3D scene graphs for perception and planning," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 5021–5028.

[39] A. Werby, C. Huang, M. Büchner, A. Valada, and W. Burgard, "Hierarchical open-vocabulary 3D scene graphs for language-grounded robot navigation," in *Proc. Robotics: Science and Systems (RSS)*, 2024.

[40] K. Rana, J. Haviland, S. Garg, J. Abou-Chakra, I. Reid, and N. Suenderhauf, "SayPlan: Grounding large language models using 3D scene graphs for scalable robot task planning," in *Proc. Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 23–72.

[41] C. Huang, O. Mees, A. Zeng, and W. Burgard, "Visual language maps for robot navigation," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 10 608–10 615. [Online]. Available: https://doi.org/10.1109/ICRA48891.2023.10160969

[42] P. Biber and T. Duckett, "Dynamic maps for long-term operation of mobile service robots," in *Proc. Robotics: Science and Systems (RSS) I*, 2005, pp. 17–24.

[43] T. Krajník, J. P. Fentanes, J. P. S. S. Santos, and T. Duckett, "FreMEn: Frequency map enhancement for long-term mobile robot autonomy in changing environments," *IEEE Transactions on Robotics*, vol. 33, no. 4, pp. 964–977, 2017.

[44] L. Schmid, M. Abate, Y. Chang, and L. Carlone, "Khronos: A unified approach for spatio-temporal metric-semantic SLAM in dynamic environments," in *Proc. Robotics: Science and Systems (RSS)*, 2024.

[45] T. Behrens, R. Zurbrügg, M. Pollefeys, Z. Bauer, and H. Blum, "Lost & found: Tracking changes from egocentric observations in 3D dynamic scene graphs," *IEEE Robotics and Automation Letters*, vol. 10, no. 4, pp. 3739–3746, 2025.

[46] X. Yin and H. Du, "GLOVE: Global verifier for LLM memory-environment realignment," 2026, arXiv preprint.

[47] G. Tang, Q. Jia, G. Chen, T. Li, Z. Huang, Z. Lv, and N. Ji, "VLM-DEWM: Dynamic external world model for verifiable and resilient vision-language planning in manufacturing," 2026, arXiv preprint.

[48] R. Bajcsy, "Active perception," *Proceedings of the IEEE*, vol. 76, no. 8, pp. 966–1005, 1988.

[49] R. Bajcsy, Y. Aloimonos, and J. K. Tsotsos, "Revisiting active perception," *Autonomous Robots*, vol. 42, no. 2, pp. 177–196, 2018.

[50] W. R. Scott, G. Roth, and J.-F. Rivest, "View planning for automated three-dimensional object reconstruction and inspection," *ACM Computing Surveys*, vol. 35, no. 1, pp. 64–96, 2003.

[51] A. Z. Ren, J. Clark, A. Dixit, M. Itkina, A. Majumdar, and D. Sadigh, "Explore until confident: Efficient exploration for embodied question answering," in *Proc. Robotics: Science and Systems (RSS)*, 2024.

[52] V. Sripada, S. Carter, F. Guerin, and A. Ghalamzan, "Scene exploration by vision-language models," 2024, arXiv preprint.

[53] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *Proc. International Conference on Learning Representations (ICLR)*, 2019.

[54] C. Michaelis, B. Mitzkus, R. Geirhos, E. Rusak, O. Bringmann, A. S. Ecker, M. Bethge, and W. Brendel, "Benchmarking robustness in object detection: Autonomous driving when winter is coming," 2019, arXiv preprint.

[55] S. Chen, J. Gu, Z. Han, Y. Ma, P. Torr, and V. Tresp, "Benchmarking robustness of adaptation methods on pre-trained vision-language models," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 36, 2023.

[56] M. C. Schiappa, S. Vyas, H. Palangi, Y. S. Rawat, and V. Vineet, "Robustness analysis of video-language models against visual and language perturbations," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 35, 2022.

[57] S. Höfer, K. Bekris, A. Handa, J. C. Gamboa, M. Mozifian, F. Golemo *et al.*, "Sim2Real in robotics and automation: Applications and challenges," *IEEE Transactions on Automation Science and Engineering*, vol. 18, no. 2, pp. 398–400, 2021.

[58] A. Avižienis, "The N-version approach to fault-tolerant software," *IEEE Transactions on Software Engineering*, vol. SE-11, no. 12, pp. 1491–1501, 1985.

[59] J. C. Knight and N. G. Leveson, "An experimental evaluation of the assumption of independence in multiversion programming," *IEEE Transactions on Software Engineering*, vol. SE-12, no. 1, pp. 96–109, 1986.

[60] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, "Improving factuality and reasoning in language models through multiagent debate," in *Proc. International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 11 733–11 763.

[61] H. Kress-Gazit, K. Hashimoto, N. Kuppuswamy, P. Shah *et al.*, "Robot learning as an empirical science: Best practices for policy evaluation," 2024. [Online]. Available: https://arxiv.org/abs/2409.09491

[62] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927. [Online]. Available: https://doi.org/10.1080/01621459.1927.10502953

[63] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947. [Online]. Available: https://doi.org/10.1007/BF02295996

[64] J. Tobin, R. Fong, A. Ray, J. Schneider *et al.*, "Domain randomization for transferring deep neural networks from simulation to the real world," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 23–30. [Online]. Available: https://doi.org/10.1109/IROS.2017.8202133

[65] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, "Robot operating system 2: Design, architecture, and uses in the wild," *Science Robotics*, vol. 7, no. 66, p. eabm6074, 2022. [Online]. Available: https://doi.org/10.1126/scirobotics.abm6074

[66] N. Koenig and A. Howard, "Design and use paradigms for Gazebo, an open-source multi-robot simulator," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, vol. 3, 2004, pp. 2149–2154. [Online]. Available: https://doi.org/10.1109/IROS.2004.1389727

[67] Qwen Team, "Qwen3.5: Towards native multimodal agents," Official blog post and Hugging Face model card (Qwen/Qwen3.5-9B, Apache-2.0), Feb. 2026, model card: https://huggingface.co/Qwen/Qwen3.5-9B. [Online]. Available: https://qwen.ai/blog?id=qwen3.5

[68] S. Bai, K. Chen, X. Liu, J. Wang *et al.*, "Qwen2.5-VL technical report," 2025. [Online]. Available: https://arxiv.org/abs/2502.13923

[69] S. Bai, Y. Cai, R. Chen, K. Chen *et al.*, "Qwen3-VL technical report," 2025. [Online]. Available: https://arxiv.org/abs/2511.21631

[70] M. Chen, "Introducing Clef: Our open-source decision models, and new RL fine-tuning platform," Cloudflare Blog, Oct. 2026. [Online]. Available: https://blog.cloudflare.com/clef-decision-models

[71] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer, "Sigmoid loss for language image pre-training," in *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 11 975–11 986.

[72] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proc. International Conference on Machine Learning (ICML)*, ser. PMLR, vol. 139, 2021, pp. 8748–8763.

[73] P. V. A. B. Venâncio, A. C. Lisboa, and A. V. Barbosa, "An automatic fire detection system based on deep convolutional neural networks for low-power, resource-constrained devices," *Neural Computing and Applications*, vol. 34, no. 18, pp. 15 349–15 368, 2022.